

TALHA GHAFOOR

QA Engineer | AI/ML Engineer

+92 349 9675845 | talhaghafoor84@gmail.com | Portfolio: portfolio-rho-one-vkx0n2h9ia.vercel.app

PROFESSIONAL SUMMARY

Software Engineering student with hands-on experience in both software quality assurance and applied AI/ML engineering — manual and automated testing (Cypress, Postman, AI-assisted automation), NLP, computer vision, and model evaluation (precision, recall, F1, AUC-ROC). Full-stack background in Python, FastAPI, React, AWS, Docker. Seeking a QA Engineer or AI/ML Engineer role.

PROFESSIONAL EXPERIENCE

AI and Machine Learning Engineering Intern

Jul 2025 - Sep 2025

Digital Empowerment Network

- Designed and executed test cases to validate a mental health detection model (TF-IDF + Logistic Regression) against defined accuracy criteria, achieving ~84% F1-score through iterative test cycles and threshold tuning.
- Identified a class-imbalance defect in a heart disease detection pipeline during test analysis; applied SMOTE as a corrective fix and verified the improvement via regression testing, raising minority-class recall from ~71% to 83%.
- Authored structured QA-style evaluation reports (accuracy, precision, recall, AUC-ROC) and presented findings to the supervising engineering team at project close-out.

AI and Machine Learning Engineering Intern

Jun 2025 - Jul 2025

DevelopersHub Corporation

- Conducted systematic testing and validation of a CNN-based image classifier across multiple training iterations, confirming ~89% validation accuracy against target benchmarks.
- Performed data quality testing on a ~6,000-image dataset — cleaning, labeling, and flagging inconsistencies — to catch defects before training and reduce overfitting.
- Documented test results and defect/error patterns across training runs in structured weekly progress reports.

KEY PROJECTS

GenMark - Multi-Modal AI Content Generation Platform (FYP)

2025 - Present

HITEC University | Python, FastAPI, React, AWS, Docker

- Wrote and executed automated end-to-end test suites for the React frontend using Cypress, covering core user flows across content generation and preview.
- Performed API testing on FastAPI backend endpoints using Postman, validating request/response contracts, status codes, and error handling across text and image generation routes.
- Applied AI-assisted automation testing to accelerate test case generation and expand test coverage across the suite.
- Designed the on-demand AI inference architecture behind the product — lazy-loaded models across AWS ECS Fargate, Lambda, and API Gateway — that these test suites validate.

Full-Stack E-Commerce Platform

2025

Personal Project | Next.js, Laravel, Supabase (PostgreSQL), Stripe, AWS S3

- Manually tested core commerce flows — product discovery, cart/checkout, and payment — across a Next.js frontend and Laravel REST API backend to catch functional defects before release.
- Validated Stripe webhook behavior to confirm accurate, real-time order-status synchronization between payment gateway and backend.

TECHNICAL SKILLS

Quality Assurance: Manual Testing, Test Case Design, Cypress (E2E/Frontend Automation), API Testing (Postman), AI-Assisted Test Automation, Regression Testing, Model/Data Evaluation (Precision, Recall, F1, AUC-ROC, SMOTE)

AI & ML: scikit-learn, NLP, Computer Vision, CNN, HuggingFace Transformers, Stable Diffusion, Qwen

Core Languages: Python, JavaScript | Familiar With: Java, C++, Dart

Web Development: React, PHP, Laravel, FastAPI, Flask, REST APIs, Tailwind CSS, HTML, CSS

Databases & Cloud: PostgreSQL (Supabase), MongoDB, MySQL | AWS (ECS Fargate, Lambda, API Gateway, S3, ECR), OCI, Docker

Tools: Git, GitHub, Postman, Cypress, Jupyter Notebook, VS Code

EDUCATION

BS Software Engineering (2022 - 2026 Expected)

HITEC University, Taxila - Dept. of Computer Science

Coursework: Software Quality Assurance, Machine Learning, Cloud Computing, Web Engineering, Database Systems

Intermediate - ICS (Computer Science) (2020 - 2022) | KRL College, Kahuta

Matriculation - Computer Science (2018 - 2020) | Mumtaz Grammar School, Wazirabad

CERTIFICATIONS

OCI 2025 Certified AI Foundations Associate, Oracle University (Aug 2025 - Aug 2027) | Certified Python & Machine Learning, Mars Computer Academy (2025)